

Homological flat bands and all-rank closed forms in strong-coupling $SU(N)$ Hamiltonian lattice gauge theory

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Abstract

We study the Kogut–Susskind Hamiltonian of pure $SU(N)$ gauge theory on a periodic cubic lattice, in strong-coupling perturbation theory on the charge-conjugation-odd one-plaquette shell. Charge-odd plaquette states are oriented 2-chains, and we prove that the second-order effective Hamiltonian is t_N times the down Laplacian $L_\downarrow = \partial_2^\dagger \partial_2$ of the cubical complex, with $t_N = 2N(N^2 - 4)/[(N^2 - 1)(2N^2 - 1)(4N^2 - 9)]$. The spectrum on each Bloch fibre is $\{0, q, q\}$; the flat band is $\ker \partial_2$, of exact dimension $L^3 + 2$ on the L^3 torus, spanned by cube boundaries and three wrapping sheets, and it touches the dispersive doublet singularly at Γ . We then prove a centre-parity theorem: for even N every odd order of the expansion vanishes identically, and for odd N nothing survives below order $N - 2$, where the only term is a one-plaquette vertex $-(N/(N + 1))^{N-3}/((N - 3)!)^2$. Hence the band is exactly flat through third order for every $N \geq 4$; for $SU(3)$ the same holds by exact computation. At fourth order, exact computation over the field $\mathbb{Q}(N)$ gives every cluster cumulant as a single rational function of N ; the kernel is a polynomial in the two Hodge Laplacians plus one further operator, which forces two of the four symmetry-allowed shape coefficients to vanish, and the band acquires the dispersion $[\alpha_N \sum_i X_i^2 + \beta_N \sum_{i < j} X_i X_j]/(2 \sum_i X_i)$, $X_i = 1 - \cos k_i$, with $\alpha_N = 640/[N(N^2 - 1)^3]$ and β_N an explicit rational function. We prove $\beta_N > \alpha_N > 0$ for all $N \geq 3$, so the minimum is at Γ , the maximum at the zone corner, and the bandwidth is $\alpha_N + \beta_N$. Finally, the one-plaquette problem is N free fermions in a cosine potential minus a centre-of-mass term; we prove that in the $U(N)$ model the fundamental doublet does not split below order $N - 1$, which turns an observed cancellation between irreducible channels into a theorem at every even order and yields the bound $O(N^{-(m+1)})$ on the order- m splitting. The observed exponent is $N^{-(2m-1)}$; we state what remains to prove it. The fourth-order results are computer assisted and rest on four assumptions that we state explicitly. All statements concern a fixed lattice spacing; none is a statement about the continuum limit.

1 Introduction

Flat bands — dispersion branches that are constant over the whole Brillouin zone — are usually engineered. They arise from destructive interference on line graphs and related frustrated geometries [10, 11, 12, 13], they carry compact localised states, and when they touch a dispersive band the touching is forced by real-space topology [14, 15]. A unifying algebraic description is that the hopping matrix is a Gram matrix $MM^\dagger$ of a rectangular incidence matrix M , and the flat band is $\ker M^\dagger$ [16, 17]. Closed two-dimensional surfaces play the same role in 2-form spin liquids [18], and a recent construction obtains extensive face-graph degeneracy from local graph topology [19].

This paper shows that the same structure appears, with no tuning, in the strong-coupling expansion of non-Abelian lattice gauge theory. In the Hamiltonian formulation [1] the lowest

excitations at strong coupling are single plaquettes of fundamental flux. The charge-conjugation-odd combination changes sign when the orientation of the plaquette is reversed, so the charge-odd one-plaquette shell is canonically the space C_2 of 2-chains of the spatial cubical complex. Group integration over the link shared by two neighbouring plaquettes then produces a hopping amplitude whose sign is the product of the two incidence numbers. The second-order effective Hamiltonian is therefore a multiple of $\partial_2^\dagger \partial_2$, and its zero modes are the 2-cycles.

1.1 Results

We separate statements *proved* in the text from statements obtained by *exact computation*, which are labelled “Computed result” and come with their assumptions.

1. **Second order (proved; Section 3).** On the charge-odd shell, $H_2 = c_2(N) I + t_N L_\downarrow$ with t_N as in the abstract (Theorem 3.2). On each Bloch fibre $L_\downarrow(k) = q(k)I - w(k)w(k)^\dagger$ with spectrum $\{0, q, q\}$ (Proposition 3.4). The flat band has dimension $L^3 + 2$ (Theorem 3.5) and its Bloch vector is discontinuous at Γ (Proposition 3.6).
2. **Odd orders (proved; Section 4).** The centre-parity theorem (Theorem 4.1) and its closed-form vertex. Corollary: the band is flat through third order for every $N \geq 4$. For $SU(3)$ the third-order operator is $\frac{1975}{124848} L_\downarrow$ plus a scalar (Computed result 4.3).
3. **Fourth order (computed, with proved consequences; Section 5).** The kernel is a Hodge polynomial plus one operator (Computed result 5.2); consequently two shape coefficients vanish (Theorem 5.4, proved from the form). All cumulants are rational functions of N and β_N is assembled from three of them (Computed result 5.5, Theorem 5.7). The band extrema and bandwidth follow (Theorem 5.9, proved from the closed forms), as does a single angular invariant governing the coupling of the flat band to the dispersive doublet (Theorem 5.10).
4. **One plaquette as free fermions (proved; Section 6).** The $U(N)$ doublet does not split below order $N - 1$ (Theorem 6.1), and on the full degenerate level its off-diagonal element is computed in closed form and equals $\frac{N-2}{4} \delta_A \delta_B (1 + O(u)) = O(u^{2N})$ (Theorem 6.2); the $SU(N)$ splitting is generated entirely by the centre-of-mass term and is $O(N^{-(m+1)})$ at order m (Corollary 6.3). The stronger observed law is Conjecture 6.4.

1.2 Relation to earlier work

Hamiltonian strong-coupling series for glueball masses were developed in [2, 3] and, in the Euclidean formulation, in [4]; they are series for energies at zero momentum. After conversion between normalisations, the Γ -point coefficients of the charge-odd shell obtained here agree through third order with the 1^{+-} series of [3]. What is new here is the momentum dependence, its identification with the Laplacian of the cell complex, and the closed forms in N . Group integrals are evaluated with the Weingarten calculus [5, 6]. The representation of one-plaquette class functions by free fermions on a circle is classical [20, 21, 22, 23]; our use of it, to prove a cancellation in perturbation theory, appears to be new. Discrete Hodge theory on cell complexes goes back to [9]. Truncated plaquette bases of this kind are also the starting point of current proposals for quantum simulation [24, 25].

1.3 What is not claimed

Every result is a statement about an effective operator of degenerate perturbation theory on a finite-dimensional shell at fixed lattice spacing and finite volume. Convergence of the expansion is not addressed beyond the isolation of the shell (Proposition 2.1), and nothing here bears on

the continuum limit: along an asymptotically free trajectory the coupling u of (2) tends to infinity, outside any strong-coupling regime.

2 Setting

2.1 Hamiltonian and normalisation

Let $\Lambda = (\mathbb{Z}/L\mathbb{Z})^3$ with $L \geq 3$, with oriented links $\ell = (x, \mu)$ carrying $U_\ell \in \text{SU}(N)$, $N \geq 3$, and Hilbert space $\bigotimes_\ell L^2(\text{SU}(N), dU_\ell)$ with Haar measure. Write C_ℓ for the quadratic Casimir operator of link ℓ , normalised so that on the fundamental representation F

$$\begin{aligned} C_2(F) = C_F = \frac{N^2 - 1}{2N}, \quad C_2(\text{Adj}) = N, \\ C_2(\Lambda^2 F) = \frac{(N+1)(N-2)}{N}, \quad C_2(\text{Sym}^2 F) = \frac{(N-1)(N+2)}{N}. \end{aligned} \quad (1)$$

We study

$$H = H_E - uW, \quad H_E = \frac{1}{2} \sum_\ell C_\ell, \quad W = \sum_p (\text{Tr } U_p + \text{Tr } U_p^\dagger), \quad (2)$$

where the sum runs over unoriented plaquettes and U_p is the ordered product of the four link variables around p . If the Kogut–Susskind Hamiltonian is written $aH_{\text{KS}} = \frac{g^2}{2} \sum_\ell C_\ell - \frac{1}{g^2} W$, then $H = aH_{\text{KS}}/g^2$ and $u = g^{-4}$. Physical states are gauge invariant. Charge conjugation $\mathcal{C} : U_\ell \mapsto U_\ell^*$ commutes with H , as do lattice translations, the cubic group, and the centre symmetry that multiplies all links crossing a fixed plane by an element of $\mathbb{Z}_N$.

2.2 The charge-odd shell

At $u = 0$ the ground state is the constant function $|0\rangle$, with $H_E|0\rangle = 0$. For an oriented plaquette p put

$$|p\rangle = \frac{1}{\sqrt{2}} (\text{Tr } U_p - \text{Tr } U_p^\dagger) |0\rangle. \quad (3)$$

These states are orthonormal by character orthogonality, have $H_E|p\rangle = 2C_F|p\rangle$ (four fundamental links at energy $C_F/2$ each), are odd under $\mathcal{C}$, and satisfy $|-p\rangle = -|p\rangle$, where $-p$ is p with reversed orientation. The last property is the point of the construction: *the map $p \mapsto |p\rangle$ identifies the span $\mathcal{P}$ of these states with the space C_2 of cellular 2-chains*, of dimension $3L^3$.

Proposition 2.1 (isolation). *In the gauge-invariant, $\mathcal{C}$ -odd sector of trivial centre charge, the spectrum of H_E in $[0, 2C_F + \gamma_N)$ is $\{2C_F\}$, with eigenspace $\mathcal{P}$, where $\gamma_N = C_F$ for $N = 3$ and $N \geq 5$, and $\gamma_4 = 5/4$.*

Proof. Eigenstates of H_E are labelled by an irreducible representation on each link, and the energy is $\frac{1}{2} \sum_\ell C_2(\rho_\ell)$; the smallest nonzero Casimir is C_F . Gauge invariance forbids a vertex with exactly one excited link, so the excited links form a subgraph with all degrees at least two. In the cubic lattice with $L \geq 3$ such a subgraph with trivial centre charge has at least four links, and with exactly four links it is a plaquette; at a two-valent vertex the two representations must be conjugate along the path, so a plaquette state is $\chi_\rho(U_p)|0\rangle$ with energy $2C_2(\rho)$. The values $\rho = F, \bar{F}$ give the shell and its $\mathcal{C}$ -even partner. The remaining plaquette states have $2C_2(\rho) - 2C_F \geq 2C_2(\Lambda^2 F) - 2C_F = (N-3)(N+1)/N$ for $N \geq 4$ and $\geq 2C_2(\text{Adj}) - 2C_F = 10/3$ for $N = 3$, where $\Lambda^2 F = \bar{F}$. There is no admissible subgraph with five links; with six or more links the energy is at least $3C_F$. A Polyakov line in a representation of zero N -ality has energy at least $3N/2 > 3C_F$. Comparing $(N-3)(N+1)/N$ with C_F gives γ_N . $\square$

By Proposition 2.1 and the boundedness of W at finite volume, analytic perturbation theory applies to the shell. We write P for the projector onto $\mathcal{P}$, $Q = 1 - P$, $\mathcal{R} = Q(2C_F - H_E)^{-1}Q$, and use the Hermitian (des Cloizeaux) effective Hamiltonian [7, 8]

$$H_{\text{eff}} = 2C_F + \sum_{m \geq 1} u^m H_m, \quad H_1 = -PWP, \quad H_2 = PWRWP, \\ H_3 = -(PWRWRWP - \tfrac{1}{2}\{PWR^2WP, PWP\}),$$

with energies measured from the perturbed vacuum energy. By translation and cubic symmetry the diagonal of every H_m is a scalar.

3 Second order: the down Laplacian

3.1 The cell complex

Let C_0, C_1, C_2, C_3 be the real vector spaces on vertices, links, plaquettes and cubes, with cell bases orthonormal, and boundary maps $\partial_2[x; \mu\nu] = [x; \mu] + [x + \hat{\mu}; \nu] - [x + \hat{\nu}; \mu] - [x; \nu]$ and $\partial_3[x] = \sum_{\mu} (-1)^{\mu-1} ([x + \hat{\mu}; \mu^c] - [x; \mu^c])$, where μ^c is the complementary pair in increasing order. Then $\partial_2\partial_3 = 0$. On C_2 define

$$L_{\downarrow} = \partial_2^{\dagger}\partial_2, \quad L_{\uparrow} = \partial_3\partial_3^{\dagger}, \quad L_{\downarrow}L_{\uparrow} = L_{\uparrow}L_{\downarrow} = 0. \quad (4)$$

The matrix element $(L_{\downarrow})_{pp'} = \sum_{\ell} [p : \ell][p' : \ell]$ is 4 for $p = p'$, and for $p \neq p'$ sharing a link it is +1 if the two plaquettes induce the same orientation on that link and -1 otherwise. We write $S_{\square} = L_{\downarrow} - 4I$ for the signed shared-link adjacency.

3.2 The hopping amplitude

Lemma 3.1 (shared-link channels). *Let $p \neq p'$ share the link ℓ , and let $\chi, \chi' \in \{\chi_F, \chi_{\bar{F}}\}$. The product $\chi(U_p)\chi'(U_{p'})|0\rangle$ has unit norm. It decomposes according to the irreducible representation ρ carried by ℓ , with $\rho \in \{\mathbf{1}, \text{Adj}\}$ if the two fluxes traverse ℓ in opposite directions and $\rho \in \{\Lambda^2 F, \text{Sym}^2 F\}$ if in the same direction. The component in channel ρ has squared norm d_{ρ}/N^2 , $d_{\rho} = \dim \rho$, and is an eigenstate of H_E with energy $3C_F + \frac{1}{2}C_2(\rho)$.*

Proof. Write $U_p = U_{\ell}A$ and $U_{p'} = U_{\ell}^{\pm 1}B$, where A and B are products of the three private links of each plaquette and are independent Haar variables. Integrating over A and B with $\int A_{ij}\overline{A_{kl}}dA = \delta_{ik}\delta_{jl}/N$ gives, for the projector Π_{ρ} acting on the two copies of the representation space of ℓ , $\|\Pi_{\rho}\chi\chi'\|^2 = N^{-2} \text{Tr} \Pi_{\rho} = d_{\rho}/N^2$; summing over ρ gives 1. The six private links are fundamental and ℓ carries ρ . $\square$

Theorem 3.2 (second order). *For every $N \geq 3$,*

$$H_2 = c_2(N)I + t_N L_{\downarrow}, \quad t_N = \frac{2N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)} > 0, \quad (5)$$

where $c_2(N)$ is a scalar.

Proof. Let $p \neq p'$. A second-order path from $|p\rangle$ to $|p'\rangle$ either annihilates p and then creates p' , through the vacuum, or creates p' and then annihilates p . For flux labels χ on p and χ' on p' , the vacuum route contributes $1/(2C_F)$ independently of (χ, χ') . In the $\mathcal{C}$ -odd combination (3) the amplitude is $\frac{1}{2}[(F \rightarrow F') + (\bar{F} \rightarrow \bar{F}') - (F \rightarrow \bar{F}') - (\bar{F} \rightarrow F')]$, so every contribution independent of (χ, χ') cancels. If p and p' share no link, the intermediate state of the second route is a product state of energy $4C_F$ and its contribution is again independent of the labels; hence H_2 connects only plaquettes that share a link.

Let p, p' share ℓ . By Lemma 3.1 the second route contributes $\sum_{\rho} w_{\rho}$ over the two channels allowed by the relative direction, with

$$w_{\rho} = \frac{d_{\rho}/N^2}{2C_F - (3C_F + \frac{1}{2}C_2(\rho))} = -\frac{d_{\rho}/N^2}{C_F + \frac{1}{2}C_2(\rho)}. \quad (6)$$

With (1),

$$\begin{aligned} w_1 &= -\frac{2}{N(N^2 - 1)}, & w_{\text{Adj}} &= -\frac{2(N^2 - 1)}{N(2N^2 - 1)}, \\ w_{\Lambda^2} &= -\frac{N - 1}{(N + 1)(2N - 3)}, & w_{\text{Sym}^2} &= -\frac{N + 1}{(N - 1)(2N + 3)}, \end{aligned}$$

and the two sums are

$$\Sigma_N^{\circ} = w_1 + w_{\text{Adj}} = -\frac{2N^3}{(N^2 - 1)(2N^2 - 1)}, \quad \Sigma_N^{\text{s}} = w_{\Lambda^2} + w_{\text{Sym}^2} = -\frac{4N(N^2 - 2)}{(N^2 - 1)(4N^2 - 9)}. \quad (7)$$

Suppose p and p' induce the same orientation on ℓ . Then $(F \rightarrow F')$ and $(\bar{F} \rightarrow \bar{F}')$ run through the same-direction channels and the two mixed amplitudes through the opposite-direction channels, so the $\mathcal{C}$ -odd amplitude is $\Sigma_N^{\text{s}} - \Sigma_N^{\circ}$. If the induced orientations are opposite the roles are exchanged and the amplitude is $\Sigma_N^{\circ} - \Sigma_N^{\text{s}}$. In both cases it equals $t_N[p : \ell][p' : \ell]$ with

$$t_N = \Sigma_N^{\text{s}} - \Sigma_N^{\circ} = \frac{2N}{N^2 - 1} \cdot \frac{N^2(4N^2 - 9) - 2(N^2 - 2)(2N^2 - 1)}{(2N^2 - 1)(4N^2 - 9)} = \frac{2N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)}.$$

Hence the off-diagonal part of H_2 is $t_N S_{\square}$; the diagonal is a scalar, and absorbing $-4t_N$ into it gives (5). $\square$

Remark 3.3. (i) $t_3 = 5/612$. (ii) The formula vanishes at $N = 2$, where χ_F is real and the shell (3) is empty. (iii) As $N \rightarrow \infty$, $t_N = \frac{1}{4}N^{-3} - \frac{1}{16}N^{-5} - \frac{77}{64}N^{-7} + O(N^{-9})$, although Σ_N° and Σ_N^{s} are each $-N^{-1} + O(N^{-3})$: one order in N^{-2} cancels between the two orientation classes. This is the first instance of the pattern discussed in Section 6.

3.3 Bloch form, flat band, and multiplicity

Order the face orbitals as (12), (13), (23) and the link orbitals as 1, 2, 3, and put $d_j = e^{ik_j} - 1$, $a_j = |d_j|^2 = 4 \sin^2(k_j/2)$, $q = a_1 + a_2 + a_3$. The coboundary $\partial_2^{\dagger}$ has Bloch matrix

$$\delta(k) = \begin{pmatrix} d_2 & -d_1 & 0 \\ d_3 & 0 & -d_1 \\ 0 & d_3 & -d_2 \end{pmatrix}, \quad w(k) = (\bar{d}_3, -\bar{d}_2, \bar{d}_1)^T, \quad (8)$$

and $w(k)$ is the Bloch matrix of ∂_3 .

Proposition 3.4. $L_{\downarrow}(k) = \delta(k)\delta(k)^{\dagger} = q(k)I_3 - w(k)w(k)^{\dagger}$ and $L_{\uparrow}(k) = w(k)w(k)^{\dagger}$, so that $L_{\downarrow}(k) + L_{\uparrow}(k) = q(k)I_3$. For $q(k) > 0$ the spectrum of $L_{\downarrow}(k)$ is $\{0, q, q\}$, with $\ker L_{\downarrow}(k) = \mathbb{C} w(k)$.

Proof. The diagonal entries of $\delta\delta^{\dagger}$ are $a_1 + a_2$, $a_1 + a_3$, $a_2 + a_3$, that is $q - |w_i|^2$. The off-diagonal entries are $d_2\bar{d}_3$, $-d_1\bar{d}_3$ and $d_1\bar{d}_2$, which equal $-w_1\bar{w}_2$, $-w_1\bar{w}_3$ and $-w_2\bar{w}_3$. Since $w^{\dagger}w = q$, the vector w is a null vector and $w^{\perp}$ is the q -eigenspace. $\square$

Theorem 3.5 (multiplicity). $\dim \ker \partial_2 = (L^3 - 1) + 3 = L^3 + 2$. The kernel is the orthogonal sum of $\text{im } \partial_3$, spanned by the boundaries of the L^3 elementary cubes subject to the single relation $\sum_x \partial_3[x] = 0$, and the three-dimensional space of harmonic 2-chains, spanned modulo boundaries by the wrapping sheets $S_{\mu} = \sum_{x_{\mu}=0} [x; \mu^c]$.

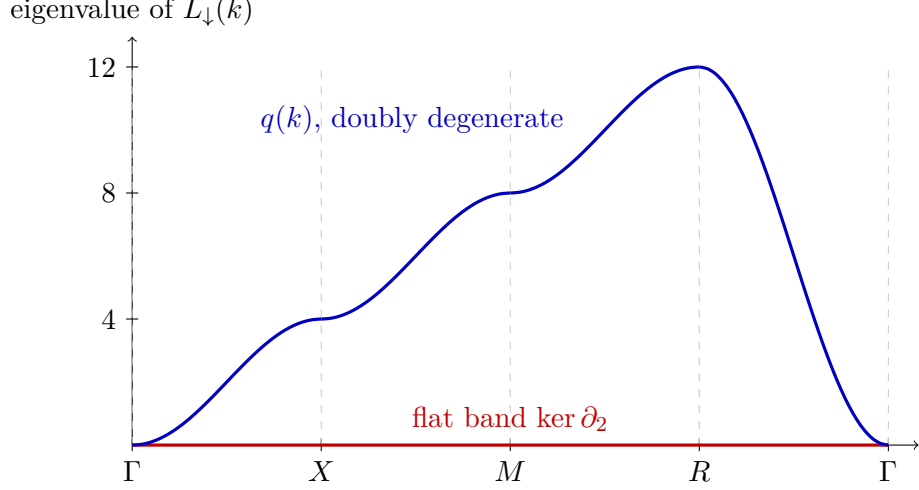


Figure 1: Spectrum of $L_{\downarrow}(k)$ along $\Gamma = (0, 0, 0)$, $X = (\pi, 0, 0)$, $M = (\pi, \pi, 0)$, $R = (\pi, \pi, \pi)$. By Theorem 3.2 the second-order bands are $c_2 u^2 + t_N u^2 \{0, q, q\}$.

Proof. By the Hodge decomposition of a finite complex [9], $\ker \partial_2 = \text{im } \partial_3 \oplus \mathcal{H}_2$ with $\mathcal{H}_2 \cong H_2(T^3; \mathbb{R})$, of dimension $b_2 = 3$. Since there are no 4-cells, $\dim \ker \partial_3 = b_3 = 1$, so $\dim \text{im } \partial_3 = L^3 - 1$; the kernel of ∂_3 is spanned by the sum of all cubes, because each plaquette lies in two cubes with opposite induced orientation. Each S_μ is a cycle and the three are independent in homology. As a check, the Bloch fibres contribute one null vector for each of the $L^3 - 1$ momenta with $q > 0$ and three at Γ , where $L_{\downarrow}(0) = 0$. $\square$

Each cube boundary is supported on six plaquettes and is annihilated by $L_{\downarrow}$: it is a compact localised state. Figure 1 shows the spectrum.

Proposition 3.6 (singular touching). *For a unit vector $\hat{n}$, $\lim_{t \rightarrow 0^+} w(t\hat{n})/\sqrt{q(t\hat{n})} = -i(n_3, -n_2, n_1)^T$. The normalised Bloch vector of the flat band therefore has no limit at Γ : the touching is singular in the sense of [15], which is the Bloch-space form of the fact that the compact localised states do not span $\ker \partial_2$.*

Proof. $\bar{d}_j = -ik_j + O(k^2)$ and $q = |k|^2 + O(k^4)$. $\square$

4 Odd orders: the centre-parity theorem

A matrix element of H_m between shell states is a sum of products of factors

$$\langle a | W \mathcal{R}^{j_1} W \dots \mathcal{R}^{j_{n-1}} W | b \rangle, \quad a, b \in \mathcal{P},$$

whose numbers n of insertions of W add up to m . Expanding W and the two shell states in traces, a factor with n insertions is a sum of Haar integrals of products of $n + 2$ face words $\text{Tr } U_p^{\pm 1}$, weighted by resolvent denominators.

Theorem 4.1 (centre parity). *Consider the expansion on $\mathbb{Z}^3$, or on the torus with L even, or on any torus for $m + 2 < L$.*

1. If N is even, $H_m = 0$ for every odd m .
2. If N is odd, $H_m = 0$ for every odd $m < N - 2$, and H_{N-2} is diagonal in the plaquette label, with

$$\langle \chi_{\bar{F}}(U_p) | H_{N-2} | \chi_F(U_p) \rangle = - \left(\frac{N}{N+1} \right)^{N-3} \frac{1}{((N-3)!)^2}. \quad (9)$$

On the $\mathcal{C}$ -odd shell $H_{N-2} = +\left(\frac{N}{N+1}\right)^{N-3}((N-3)!)^{-2} I$.

Proof. Invariance of Haar measure under $U_\ell \mapsto zU_\ell$, $z \in \mathbb{Z}_N$, shows that an integral of face words vanishes unless the net flux f_ℓ (forward minus backward traversals) is 0 modulo N on every link. Let T be the set of x -links at even y , y -links at even z and z -links at even x . Every plaquette meets T in an odd number of links: an xy -plaquette contains two x -links at consecutive values of y , exactly one of which is in T , and two y -links at the same z ; the other planes are alike. If n_ℓ is the total number of traversals of ℓ , then $n_\ell \equiv f_\ell$ modulo 2, and

$$\#\{\text{face words}\} \equiv \sum_{\ell \in T} n_\ell \equiv \sum_{\ell \in T} f_\ell \pmod{2}.$$

(1) If N is even every f_ℓ is even, so a nonvanishing factor has an even number $n+2$ of face words. If m is odd, every term of H_m contains a factor with n odd.

(2) Let N be odd and consider a nonvanishing factor with n odd. Some $\ell \in T$ has f_ℓ odd, hence $f_\ell \neq 0$, hence $|f_\ell| \geq N$, so $n+2 \geq N$. Thus for $m < N-2$ every term of H_m contains a vanishing factor. For $m = N-2$ the only possibility is a single factor with $n = N-2$, and its N face words all traverse ℓ in the same direction. Two distinct plaquettes share at most one link, so any other link of a plaquette occurring among the words is traversed only by the words of that plaquette, whose number must be 0 modulo N and is at most N . Hence all N words are the same oriented plaquette p , and the element is $\langle \chi_{\bar{F}}(U_p) | (-W) \mathcal{R}(-W) \cdots \mathcal{R}(-W) | \chi_F(U_p) \rangle$ with $N-2$ insertions of $\chi_F(U_p)$. One-plaquette states are characters, multiplication by χ_F adds a box, and the only chain of diagrams from F to the column $\bar{F} = \Lambda^{N-1} F$ is $F \rightarrow \Lambda^2 F \rightarrow \cdots \rightarrow \Lambda^{N-1} F$, each step with multiplicity one. The energy of $\chi_{\Lambda^k}(U_p) | 0 \rangle$ is $2C_2(\Lambda^k F) = k(N-k)(N+1)/N$, so $E_1 - E_k = -(k-1)(N-k-1)(N+1)/N$ and

$$(-1)^{N-2} \prod_{k=2}^{N-2} \frac{1}{E_1 - E_k} = -\left(\frac{N}{N+1}\right)^{N-3} \prod_{k=2}^{N-2} \frac{1}{(k-1)(N-k-1)} = -\left(\frac{N}{N+1}\right)^{N-3} \frac{1}{((N-3)!)^2}.$$

The $\mathcal{C}$ -odd diagonal element of an operator exchanging χ_F and $\chi_{\bar{F}}$ with amplitude v is $-v$. $\square$

The vertex is -1 at $N = 3$, which is the first-order term $H_1 = +I$ on the shell, and $-25/144$ at $N = 5$. Odd orders are therefore not the odd members of a series uniform in N : they are determinant effects, present only when N face words can close through ε -tensors.

Corollary 4.2 (flatness through third order). *For every $N \geq 4$, H_1 and H_3 are scalars on the shell ($H_1 = 0$; $H_3 = 0$ except for $N = 5$, where it is $\frac{25}{144}I$). Hence through third order the bands are those of Figure 1 and $\ker \partial_2$ is exactly flat.*

Computed result 4.3 ($\text{SU}(3)$, third order). For $N = 3$, exact evaluation of H_3 gives

$$H_{\text{eff}} = E_{\text{flat}}(u) I + \left(\frac{5}{612} u^2 + \frac{1975}{124848} u^3 \right) L_{\downarrow} + O(u^4), \quad E_{\text{flat}} = \frac{8}{3} + u + \frac{11}{306} u^2 - \frac{109151}{249696} u^3, \quad (10)$$

so the band is flat through third order for $\text{SU}(3)$ as well. The third-order operator was evaluated with two independent implementations, and its one-plaquette contributions with a third, character-based computation; the Γ -point coefficients agree with the series of [3].

5 Fourth order

5.1 Assumptions

The fourth-order statements are exact rational computations. They rest on the following, which we state once.

Assumption 5.1. (A1) Since $PWP = 0$ for $N \geq 4$, the Hermitian fourth-order operator is $H_4 = PWRWRWRWP - \frac{1}{2}\{PWR^2WP, PWRWP\}$. For $N = 3$ the terms containing PWP are included.

- (A2) On the words reached at fourth order, the eigenspace of H_E at energy $2C_F$ is exactly $\mathcal{P}$. This was verified on all 224 components encountered; it is not derived.
- (A3) The connected-cluster decomposition is exhausted by the list: pairs, two-hop chains, fans, corners, and the two cube completions. This rests on a counting argument. The pair cluster is not computed over $\mathbb{Q}(N)$; its cancellation from the off-axis coefficient rests on a parity argument that we have not formalised.
- (A4) For the closed forms in N , resolvents are taken at generic rank. The specialisation to integer N is justified by argument for $N \geq 5$ (Proposition 5.6); at $N = 3$ and $N = 4$ the closed forms are regular and agree with direct computation at that rank, as a checked fact.

All all-rank computations were performed with one implementation, a Wilson-loop calculus over $\mathbb{Q}(N)$ with Weingarten functions as symbolic Gram inverses; at $N = 3$ the central values were confirmed by two further independent implementations. An independent recomputation of the corner cumulant at some $N \geq 5$ is the most useful outstanding check.

5.2 The kernel is a Hodge polynomial plus one operator

Let R be the cross-plane part of $S_\square$: on a Bloch fibre, $R_{ij} = -w_i \bar{w}_j$ for $i \neq j$ and $R_{ii} = 0$.

Computed result 5.2 (Hodge form). For $N = 3$, on all 189 translation classes of matrix elements of H_4 ,

$$H_4 = -\tilde{\nu} (L_\uparrow - 2) + \omega S_\square^2 - \tilde{\pi} S_\square + \tilde{\sigma} I - 2C_{\text{shp}} R, \quad \tilde{\nu} = -\frac{5}{48}, \quad (11)$$

exactly, where ω is the two-hop weight of Section 5.3 and $\tilde{\pi}, \tilde{\sigma}$ are rational numbers. In particular the two-hop sector enters as the *square* of the second-order operator, and R is the only operator outside the algebra generated by $L_\downarrow$ and $L_\uparrow$. The on-site amplitude $\tilde{\sigma}$ has not been checked independently; it enters only the constant ϵ_Γ below.

Proposition 5.3 (the complement of the flat band). *On a fibre with $q > 0$ let $\psi = w$. Then $\mathbb{C}^3 = \ker L_\downarrow \oplus \ker L_\uparrow$, the projector onto $\psi^\perp$ is $Q_\psi = L_\downarrow/q$, and $Q_\psi L_\downarrow \psi = Q_\psi L_\uparrow \psi = 0$. Thus no polynomial in $L_\downarrow$ and $L_\uparrow$ couples the flat band to the dispersive doublet; in (11) only R does.*

Proof. Proposition 3.4 gives $L_\downarrow \psi = 0$, $L_\uparrow \psi = q\psi$, and $L_\uparrow = 0$ on $\psi^\perp$. $\square$

The most general energy of the flat band allowed by cubic symmetry at this order is, with $e_2 = \sum_{i < j} a_i a_j$ and $e_3 = a_1 a_2 a_3$,

$$q \varepsilon_4(k) = \epsilon_\Gamma q + A_{\text{shp}} q^2 + 4C_{\text{shp}} e_2 + B_{\text{shp}} q e_2 + D_{\text{shp}} e_3, \quad \varepsilon_4 = \frac{\langle \psi, H_4 \psi \rangle}{\langle \psi, \psi \rangle}. \quad (12)$$

Theorem 5.4 (collapse of the cubic tier). *Let $\sigma(X) = \langle \psi, X \psi \rangle$. Then*

$$\sigma(I) = q, \quad \sigma(S_\square) = -4q, \quad \sigma(S_\square^2) = 16q, \quad \sigma(L_\uparrow) = q^2, \quad \sigma(R) = -2e_2. \quad (13)$$

Consequently any operator of the form (11) has

$$B_{\text{shp}} = D_{\text{shp}} = 0, \quad A_{\text{shp}} = -\tilde{\nu}, \quad \epsilon_\Gamma = \tilde{\sigma} + 2\tilde{\nu} + 16\omega + 4\tilde{\pi}. \quad (14)$$

Moreover $\sigma(R^2) = q e_2 + 3e_3$ and $\sigma(L_\uparrow R) = -2q e_2$.

Proof. The first four symbols follow from $L_\downarrow\psi = 0$ and $L_\uparrow\psi = q\psi$. For R , $\sigma(R) = -\sum_{i \neq j} |w_i|^2 |w_j|^2 = -2e_2$. Next $(R\psi)_i = -w_i(q - a_i)$, so $\sigma(R^2) = \sum_i a_i \sum_{j \neq i} a_j (q - a_j) = 2qe_2 - (qe_2 - 3e_3)$, and $\sigma(L_\uparrow R) = q\sigma(R)$. $\square$

The last two identities delimit the mechanism. The vanishing of B_{shp} and D_{shp} is not a consequence of the number of insertions of R : the single- R operator $L_\uparrow R$ would populate B_{shp} . It is a consequence of the fact that $L_\uparrow R$ does not occur at fourth order. If R^2 occurs at sixth order, it contributes to the cubic tier in the fixed ratio $D_{\text{shp}} = 3B_{\text{shp}}$.

5.3 Closed forms over $\mathbb{Q}(N)$

Computed result 5.5 (cumulants). Computed over $\mathbb{Q}(N)$, every connected fourth-order cumulant in both $\mathcal{C}$ sectors is a single rational function of N (the pair cluster excepted, see Theorem 5.7). The two-hop weight, the single-contact dressing and the corner dressing are

$$\omega_N = \frac{N^3(N^2 - 4)^2 p_{14}(N)}{2(N^2 - 1)^3(4N^2 - 9)^3(9N^2 - 16)(2N^2 - 3)(2N^2 - 1)^3(3N^2 - 2) r_2(N)}, \quad (15)$$

$$d_N^s = \frac{2N^3(N^2 - 4)(10N^2 - 13)}{(N^2 - 1)^3(4N^2 - 9)^2(2N^2 - 1)^2}, \quad (16)$$

$$d_N^c = \frac{16N^3(N^2 - 4) p_{18}(N)}{(N^2 - 1)^3(4N^2 - 9)^3(4N^2 - 1)(9N^2 - 25)(2N^2 - 1)^3(3N^2 - 1)(4N^2 - 5) r_4(N)}, \quad (17)$$

with $r_2(N) = (4N^2 - N - 4)(4N^2 + N - 4)$, $r_4(N) = (4N^2 - 2N - 5)(4N^2 + 2N - 5)$ and p_{14}, p_{18} as in Appendix A; the cube completions between opposite and between adjacent faces are

$$\kappa_N^{\text{opp}} = -\frac{160}{N(N^2 - 1)^3}, \quad \kappa_N^{\text{adj}} = -\frac{106}{N(N^2 - 1)^3}. \quad (18)$$

Each form specialises to the value computed directly at rank N for every $N = 3, \dots, 70$. Every denominator factor is the energy denominator of an identified intermediate state; all 21,556 resolvent eigencomponents were verified to be exact eigenvectors of every single-link Casimir over $\mathbb{Q}(N)$.

Proposition 5.6 (specialisation). *The symbolic and fixed-rank computations can differ only through rank-dependent selection rules, the Weingarten function, and the spectral decomposition. No word at fourth order carries flux of magnitude at least 5 on every excited link, and every Haar integral involves at most three pairs of matrix elements per link; so for $N \geq 5$ the selection rules and the Weingarten inverse are the specialisations of the symbolic ones. The resolvent denominators have exactly one integer root with $N \geq 4$: at $N = 4$, where the state with three fundamental fluxes on one plaquette becomes $\bar{F}$ and is degenerate with the shell. Hence for $N \geq 5$ the closed forms are the fixed-rank values by argument.*

The adjacent-face cube completion deserves a remark because it is the easiest term to get wrong. The four remaining faces of the cube can be inserted in $4! = 24$ orders. In units of $1/1536$ at $N = 3$, the orders have weights -8 (eight orders), -4 (six), -2 (eight) and -1 (two), with total $-106/1536 = -53/768$, in agreement with (18). An enumeration that retains only some of the orders produces a wrong off-axis coefficient while leaving all axial and Γ -point quantities unchanged, because $e_2 = 0$ on the axes; off-axis checks are therefore essential. (An earlier enumeration in this project held 8 of the 24 orders and gave C_{shp} too negative by $25/1024$.)

Put $\beta_N = 8A_{\text{shp}} + 16C_{\text{shp}}$. In the $\mathcal{C}$ -odd sector the computed coplanar dressings are the exact negatives of the perpendicular ones, the fan clusters cancel, and under (A3) the assembly of the off-axis coefficient from the cumulants of Computed result 5.5 reduces to the left side of (19).

Theorem 5.7 (β_N from three cumulants). *As rational functions of N ,*

$$-16\omega_N + 32d_N^s - 16d_N^c + \frac{848}{N(N^2-1)^3} = \frac{P_{17}(N^2)}{N R_{20}(N^2)}, \quad (19)$$

with P_{17} and R_{20} as in Appendix A; under (A3) this is β_N .

Proof. After clearing denominators this is a polynomial identity in N . It has been checked in the Lean 4 proof assistant (by the `ring` tactic) and, independently, in exact arithmetic at the integers $N = 3, \dots, 200$, more points than the degree of the cleared identity. $\square$

For $N = 3$,

$$\omega_3 = \frac{360421351}{40327601932800}, \quad C_{\text{shp}} = -\frac{13035490122347}{550663802582400}, \quad \beta_3 = \frac{15644916262153}{34416487661400},$$

that is, $C_{\text{shp}} = -0.0236723\dots$ and $\beta_3 = 0.4545762\dots$

5.4 Dispersion, extrema and bandwidth

Put $X_i = 1 - \cos k_i = a_i/2 \in [0, 2]$ and

$$\alpha_N = 4A_{\text{shp}} = -4\kappa_N^{\text{opp}} = \frac{640}{N(N^2-1)^3}. \quad (20)$$

By (12) and (14),

$$\varepsilon_4(k) - \epsilon_\Gamma = A_{\text{shp}} q + \frac{4C_{\text{shp}} e_2}{q} = \frac{\alpha_N \sum_i X_i^2 + \beta_N \sum_{i<j} X_i X_j}{2 \sum_i X_i}. \quad (21)$$

Lemma 5.8. *For every integer $N \geq 3$, $\beta_N > \alpha_N > 0$. As $N \rightarrow \infty$, $N^7 \alpha_N \rightarrow 640$ and $N^7 \beta_N \rightarrow 6170/9$.*

Proof. All real roots of R_{20} are at most $25/9$, so $R_{20}(z) > 0$ for $z \geq 9$. The numerator of $NR_{20}(N^2)(\beta_N - \alpha_N)$ is $G(z) = P_{17}(z) - 640R_{20}(z)/(z-1)^3$ at $z = N^2$, a polynomial of degree 17 with positive leading coefficient; isolating its real roots in exact arithmetic shows that all are smaller than 9. The same holds for P_{17} , whose eleven real roots lie in $(0, 2.19)$. The limits are ratios of leading coefficients. $\square$

Theorem 5.9 (extrema). *Assume the form (21). For every $N \geq 3$ the function $\varepsilon_4 - \epsilon_\Gamma$ is positive for $k \neq \Gamma$, tends to 0 as $k \rightarrow \Gamma$, and attains its maximum exactly at $R = (\pi, \pi, \pi)$, where it equals $\alpha_N + \beta_N$. On a coordinate axis it equals $\alpha_N X/2$. The fourth-order bandwidth of the flat band is therefore*

$$\Delta_4(N) = (\alpha_N + \beta_N) u^4, \quad \Delta_4(3) = \frac{29985119454403}{34416487661400} u^4.$$

Proof. Write $f = (\alpha \sum X_i^2 + \beta e)/(2s)$ with $s = \sum X_i$, $e = \sum_{i<j} X_i X_j$. With $t = X_2 + X_3$,

$$2s^2 \frac{\partial f}{\partial X_1} = \alpha X_1^2 + 2\alpha X_1 t + (\beta - \alpha)(X_2^2 + X_3^2) + \beta X_2 X_3,$$

which by Lemma 5.8 is positive on $[0, 2]^3 \setminus \{0\}$. So f is strictly increasing in each variable and its maximum on the cube is attained only at $X = (2, 2, 2)$, where $f = (12\alpha + 12\beta)/12$. Positivity of f and its limit at the origin are evident. $\square$

For $N = 3$ the theorem rests on Computed result 5.2. For $N \geq 4$ the form (21) rests on (A3) and on the vanishing of B_{shp} and D_{shp} , which has been verified by direct computation at $N = 4, 5, 6$ and at the larger ranks examined.

5.5 Coupling to the dispersive doublet

Theorem 5.10 (one angular invariant). *Let $p = \psi/\sqrt{q}$ and $x_i = a_i/q$. For H_4 of the form (11),*

$$\|Q_\psi H_4 p\|^2 = 4C_{\text{shp}}^2 q^2 V(x), \quad V(x) = \sum_i x_i^3 - \left(\sum_i x_i^2\right)^2 = \sum_{i < j} x_i x_j (x_i - x_j)^2 \geq 0. \quad (22)$$

V vanishes exactly when the nonzero x_i are equal: on the axes, the face diagonals and the body diagonals, where the flat-band vector is an exact eigenvector of H_4 .

Proof. By Proposition 5.3, $Q_\psi H_4 p = -2C_{\text{shp}} Q_\psi R p$. From $(R\psi)_i = -w_i(q - a_i)$ and $\sigma(R) = -2e_2$, $(Q_\psi R\psi)_i = w_i(a_i - q + 2e_2/q) = q w_i(x_i - \sum_j x_j^2)$, using $2e_2 = q^2(1 - \sum_j x_j^2)$. Hence $\|Q_\psi R\psi\|^2 = q^3 \sum_i x_i (x_i - \sum_j x_j^2)^2 = q^3 V(x)$. $\square$

Since the doublet lies $t_N u^2 q$ above the flat band, second-order perturbation theory in this coupling contributes $-(4C_{\text{shp}}^2/t_N) q V(x) u^6$ to the flat-band energy; for $N = 3$ the coefficient is $-0.2743614\dots$. This is one contribution to the sixth order, not the sixth order.

6 One plaquette as free fermions

6.1 The representation

Gauge-invariant functions of a single plaquette variable are class functions. After multiplication by the Vandermonde determinant they become antisymmetric functions of the eigenvalue angles $\theta_1, \dots, \theta_N$, and characters become Slater determinants $|p_1, \dots, p_N\rangle$ of plane waves with momenta $p_j \in \mathbb{Z} + \frac{N-1}{2}$ [20, 21, 22, 23]. The trivial representation is the Fermi sea $\{-\frac{N-1}{2}, \dots, \frac{N-1}{2}\}$; a representation with highest weight λ has $p_j = \lambda_j - j + \frac{N+1}{2}$. The Casimir formula reads

$$2C_2(\lambda) = \sum_j p_j^2 - \frac{P^2}{N} - E_{\text{sea}}, \quad P = \sum_j p_j, \quad E_{\text{sea}} = \frac{N(N^2 - 1)}{12}, \quad (23)$$

and multiplication by χ_F (respectively $\chi_{\bar{F}}$) raises (lowers) the momentum of one fermion by one. For a single plaquette, whose four links carry the same representation, the Hamiltonian (2) becomes

$$H_{\text{SU}(N)} = H_{\text{U}(N)} - \frac{P^2}{N}, \quad H_{\text{U}(N)} = \sum_{j=1}^N (p_j^2 - 2u \cos \theta_j) - E_{\text{sea}}, \quad (24)$$

that is, N free fermions in a cosine potential with the centre-of-mass kinetic energy removed. The doublet is F (sea with $\frac{N-1}{2} \rightarrow \frac{N+1}{2}$, $P = 1$) and $\bar{F}$ (sea with $-\frac{N-1}{2} \rightarrow -\frac{N+1}{2}$, $P = -1$). For $\text{SU}(N)$, Slater determinants that differ by a common shift of all momenta are identified ($\det U = 1$), and (24) is invariant under the shift. The shifted copy of $\bar{F}$ with $P = N - 1$ is $N - 2$ steps from F ; this is the vertex of Theorem 4.1, and below order $N - 2$ the identification plays no role. Let $K = \sum_m u^m K_m$ be the effective Hamiltonian on the span of F and $\bar{F}$. The splitting between the $\mathcal{C}$ -odd and $\mathcal{C}$ -even one-plaquette energies is

$$S(u) = \sum_m S_m u^m, \quad S_m = -2 K_m[F][\bar{F}]. \quad (25)$$

6.2 The $U(N)$ doublet does not split

Theorem 6.1. *For the Hamiltonian $H_{U(N)}$ of (24) the expansion of K is well defined through order $N-2$, and $K_m = \kappa_m I$ with a scalar κ_m for every $m \leq N-2$. In particular $K_m[F][\bar{F}] = 0$ and $S_m = 0$ for $m \leq N-2$.*

Proof. $H_{U(N)}$ is a sum of copies of the one-body operator $h = p^2 - u(T + T^\dagger)$, where T raises the momentum by one, so its eigenvalues on antisymmetric states are sums of N distinct eigenvalues of h . At $u = 0$ the levels p^2 of h are degenerate in pairs $\{p, -p\}$, and h commutes with the reflection $p \mapsto -p$. The effective operator of h on such a pair is therefore of the form $\begin{pmatrix} a & b \\ b & a \end{pmatrix}$, and the coefficient of u^m in b is a sum over walks of length m from p to $-p$ with steps ± 1 , which requires $m \geq 2|p|$. Hence the two eigenvalues emanating from the pair satisfy $e_p^+ - e_p^- = O(u^{2|p|})$.

At $u = 0$ the level of F is fourfold degenerate: a particle-hole excitation of the sea costs at least N , with equality only when a fermion moves from $\pm \frac{N-1}{2}$ to $\pm \frac{N+1}{2}$ (either sign for each), so the level consists of F , $\bar{F}$ and the two states $D_\pm$ of total momentum $\pm N$. All four occupy every pair with $|p| \leq \frac{N-3}{2}$ fully and one member each of the pairs $\mathbf{A} = \{\pm \frac{N-1}{2}\}$ and $\mathbf{B} = \{\pm \frac{N+1}{2}\}$. The four exact eigenvalues emanating from this level are therefore $E_{\text{in}}(u) + e_{\mathbf{A}}^\pm + e_{\mathbf{B}}^\pm$, where E_{in} is the energy of the fully occupied inner pairs, and they all lie within $O(u^{N-1})$ of their mean $E_0(u)$. The Hermitian effective Hamiltonian $K^{(4)}$ on the fourfold level is analytic in u and has these eigenvalues, so $\|K^{(4)} - E_0(u)I\| = O(u^{N-1})$ and every Taylor coefficient of $K^{(4)} - E_0 I$ of order at most $N-2$ vanishes. Finally, $D_\pm$ is at least $N-1$ steps from F and from $\bar{F}$, so through order $N-2$ the states $D_\pm$ do not occur among the intermediate states of the expansion on $\{F, \bar{F}\}$, and K_m is the corresponding block of $K_m^{(4)}$. $\square$

On the fourfold level, which is a well-defined model space at every order, much more is true, and the element can be computed in closed form. Let $h^\pm$ be the restrictions of h to even and odd functions of p ; in the basis $(|k\rangle \pm |-k\rangle)/\sqrt{2}$, $k > 0$ (together with $|0\rangle$ in the even sector when N is odd), they are Jacobi matrices on a half line which differ only at the boundary. Write $e_p^\pm$ for their eigenvalues, $\delta_p = e_p^+ - e_p^-$, $p_{\mathbf{A}} = \frac{N-1}{2}$, $p_{\mathbf{B}} = \frac{N+1}{2}$, and

$$X^\pm = \sum_{|p| < p_{\mathbf{A}}} e_p^\pm - \tau^\pm, \quad x^\pm = X^\pm + e_{\mathbf{A}}^\pm, \quad Y^\pm = X^\pm + e_{\mathbf{B}}^\pm, \quad (26)$$

where the sum runs over the eigenvalues of $h^\pm$ emanating from the inner sites and $\tau^\pm$ is the trace of $h^\pm$ over the sites $k \leq p_{\mathbf{A}}$. Let $K^{\mathbf{B}} = W \text{diag}(E) W^{-1}$ be Bloch's effective Hamiltonian on the fourfold level, where W is the matrix of components of the four exact eigenvectors on the level; it has the same eigenvalues as $K^{(4)}$.

Theorem 6.2 (the doublet element in closed form). *For every $N \geq 3$, with χ_F and $\chi_{\bar{F}}$ normalised as characters,*

$$\langle F | K^{\mathbf{B}} | \bar{F} \rangle = \frac{\delta_{\mathbf{A}} - \delta_{\mathbf{B}}}{4} \cdot \frac{r_1 - r_2}{1 - r_1 r_2}, \quad r_1 = \frac{x^+}{Y^-}, \quad r_2 = \frac{x^-}{Y^+}, \quad (27)$$

exactly, and

$$r_1 - r_2 = -\frac{\bar{x}(\delta_{\mathbf{A}} + \delta_{>}) + \bar{Y}(\delta_{\mathbf{B}} + \delta_{>})}{Y^+ Y^-}, \quad \delta_{>} = \sum_{p > p_{\mathbf{B}}} \delta_p, \quad (28)$$

with $\bar{x} = \frac{1}{2}(x^+ + x^-)$, $\bar{Y} = \frac{1}{2}(Y^+ + Y^-)$. Consequently

$$\langle F | K^{\mathbf{B}} | \bar{F} \rangle = \frac{N-2}{4} \delta_{\mathbf{A}} \delta_{\mathbf{B}} (1 + O(u)) = O(u^{2N}), \quad (29)$$

so $K_m^{\mathbf{B}}[F][\bar{F}] = 0$ for every $m \leq 2N-1$.

Proof. Step 1: reduction to a 2×2 block. Label the model states by $(a, b) \in \{\pm\}^2$: the inner sea together with momenta $a p_A$ and $b p_B$, in this order. Then $F = (-, +)$, and the state $(+, -)$ is $-\bar{F}$ in the character normalisation, because reflection reverses the order of the N momenta of the sea and of the $N - 2$ inner momenta, and the two signs have product -1 . The exact states are $\Phi_{\text{in}} \wedge \phi_A^\sigma \wedge \phi_B^\tau$ with energies $E_{\sigma\tau} = E_{\text{in}} + e_A^\sigma + e_B^\tau$. Expanding the $N \times N$ overlap determinant along the inner block gives

$$W_{(a,b),(\sigma\tau)} \propto g(ap_A, \phi_A^\sigma) g(bp_B, \phi_B^\tau) - g(ap_A, \phi_B^\tau) g(bp_B, \phi_A^\sigma),$$

where $g(k, \phi)$ is the minor with rows the inner momenta and k , and columns the inner orbitals and ϕ , and the constant is common to all entries. The inner spaces are reflection invariant, so $g(-k, \phi^\sigma) = \sigma g(k, \phi^\sigma)$. Regard the model space as functions of (a, b) with basis $1, a, b, ab$. The column $(+, +)$ is then proportional to 1 and $(-, -)$ to ab , while $(-, +)$ and $(+, -)$ are $y_2 a - z_2 b$ and $y_1 b - z_1 a$ with

$$\begin{aligned} y_1 &= g(p_A, \phi_A^+) g(p_B, \phi_B^-), & z_1 &= g(p_A, \phi_B^-) g(p_B, \phi_A^+), \\ y_2 &= g(p_A, \phi_A^-) g(p_B, \phi_B^+), & z_2 &= g(p_A, \phi_B^+) g(p_B, \phi_A^-). \end{aligned}$$

So K^B has eigenvectors 1 and ab and a 2×2 block on $\{a, b\}$ with eigenvalues E_{-+}, E_{+-} . The indicator functions of $(-, +)$ and $(+, -)$ are $\frac{1}{4}(1 \mp a \pm b - ab)$, whence

$$\langle (-, +) | K^B | (+, -) \rangle = \frac{1}{4} [E_{++} + E_{--} - E_{-+} - E_{+-}] + \frac{1}{4} (K_{ab} + K_{ba}).$$

The first bracket vanishes identically because the energies are additive. The second is computed from the block: with $r_i = z_i/y_i$, $K_{ab} + K_{ba} = (E_{+-} - E_{-+})(r_2 - r_1)/(1 - r_1 r_2)$, and $E_{+-} - E_{-+} = \delta_A - \delta_B$.

Step 2: the minors. Passing to reflection-adapted rows, $g(k, \phi^\sigma)$ is, up to a factor independent of k , the minor $M^\sigma[k; \phi]$ of the eigenvector matrix of the Jacobi matrix h^σ with rows the inner sites and k . An eigenvector of a Jacobi matrix with eigenvalue e has components $c(0)\pi_i(e)$, where π_i is a polynomial of degree i and $\pi_i(e) = \det(e - J_{[0,i]})/(b_0 \cdots b_{i-1})$. If the inner sites are $0, \dots, n-1$, then p_A is site n and p_B is site $n+1$, the coupling between them is $-u$, and both minors are generalised Vandermonde determinants:

$$\frac{M^\sigma[p_B; \phi]}{M^\sigma[p_A; \phi]} = -\frac{1}{u} \left(\sum_{\text{inner}} e^\sigma + e_\phi - \tau^\sigma \right).$$

Hence $r_1 = x^+/Y^-$ and $r_2 = x^-/Y^+$, which proves (27) after the sign of Step 1.

Step 3: the trace identity. Since $h^+ - h^-$ is supported at the boundary, inside the sites counted by $\tau^\pm$, one has $\sum_p \delta_p = \tau^+ - \tau^-$ (with the unpaired level $p = 0$ included in the sum when N is odd), so $X^+ - X^- = -(\delta_A + \delta_B + \delta_>)$. Therefore $x^+ - x^- = -(\delta_B + \delta_>)$ and $Y^+ - Y^- = -(\delta_A + \delta_>)$, and $r_1 - r_2 = (x^+ Y^+ - x^- Y^-)/(Y^+ Y^-)$ gives (28).

Step 4: leading order. The quantity $x^\pm$ is the sum of the lowest $n+1$ eigenvalues of $h^\pm$ minus the trace of the corresponding block, which at second order receives only the contribution $u^2/(p_A^2 - p_B^2)$ of the bond between p_A and p_B : $\bar{x} = -u^2/N + O(u^3)$, and $\bar{Y} = N + O(u^2)$. The splitting of the pair $\pm p$ is twice the coupling generated by the shortest walk, $\delta_p = 2(-u)^{2p}/((2p-1)!)^2 (1 + O(u^2))$, so $u^2 \delta_A = N^2(N-1)^2 \delta_B(1 + O(u^2))$ and $\delta_> = O(u^{N+3})$. Then $-\bar{x} \delta_A - \bar{Y} \delta_B = N \delta_B [(N-1)^2 - 1] (1 + O(u))$, so $r_1 - r_2 = (N-2) \delta_B (1 + O(u))$, and (29) follows since $\delta_B = O(u^2 \delta_A)$ and $r_1 r_2 = O(u^4)$. $\square$

The identity (27) has been checked against a direct evaluation of $W \text{diag}(E) W^{-1}$ to more than fifty digits for $N = 3, \dots, 7$ (check_doublet_2N.py). The mechanism is transparent in Step 1: if the overlaps factorised over the two labels, K^B would be a sum of single-label flips and the element would vanish identically, because F and $\bar{F}$ differ in both labels; the exchange term of the 2×2 minor is what survives, and it is proportional to the *smaller* splitting.

Paths from F to $\bar{F}$ do exist at low order; they cancel. At second order there are two, through the sea and through the adjoint (both excitations present), with denominators $+N$ and $-N$:

$$K_2^{\text{U}(N)}[F][\bar{F}] = \frac{1}{N} - \frac{1}{N} = 0, \quad K_2^{\text{SU}(N)}[F][\bar{F}] = \frac{1}{N - \frac{1}{N}} - \frac{1}{N + \frac{1}{N}} = \frac{2N}{N^4 - 1}, \quad (30)$$

so that $S_2 = -4N/(N^4 - 1) = -4N^{-3} + O(N^{-7})$ for $\text{SU}(N)$ with $N \geq 5$.

6.3 Consequences for $\text{SU}(N)$

By (23) the $\text{SU}(N)$ problem has the same states and matrix elements, and each denominator $E_F - E_i$ of the $\text{U}(N)$ expansion is replaced by $E_F - E_i + (P_i^2 - 1)/N$, where P_i is the momentum of the intermediate state.

Corollary 6.3. *Let m be even and $N \geq m + 3$.*

1. $S_m^{\text{SU}(N)}$ is exactly the difference between the expansion evaluated with denominators $E_F - E_i + (P_i^2 - 1)/N$ and with $E_F - E_i$: the splitting is generated entirely by the centre-of-mass term.
2. Decompose S_m according to the first intermediate state. Every channel is $O(N^{-(m-1)})$, and the leading coefficients of the channels sum to zero.
3. $S_m^{\text{SU}(N)} = O(N^{-(m+1)})$.

Proof. For $N \geq m + 3$ no determinant path contributes, so the $\text{SU}(N)$ expansion is the $\text{U}(N)$ expansion with shifted denominators, and (1) is Theorem 6.1. An intermediate state with n boxes and antiboxes has $E_i - E_F = (n - 1)N + O(1)$ by (23), and the states with $n = 1$ are F and $\bar{F}$ themselves, which are excluded; so each of the $m - 1$ denominators of a term is of size N and each channel is $O(N^{-(m-1)})$. The number of terms is independent of N for $N \geq m + 3$. The centre-of-mass correction is a relative $O(N^{-2})$ change of each denominator, so the leading coefficient of each channel is its $\text{U}(N)$ value, and these sum to zero by Theorem 6.1; this proves (2) and (3). $\square$

Exact computation of S_m over $\mathbb{Q}(N)$ for even $m \leq 10$ shows the following. For $4 \leq m \leq 10$ exactly four channels contribute — the singlet, the adjoint, and the two two-box states — with leading coefficients $(-4, +2, +1, +1) \cdot 2^{m/2-2}$; at $m = 2$ only the singlet and adjoint occur, with $(-2, +2)$ as in (30). Corollary 6.3(2) explains why these sum to zero. The computation shows more:

Conjecture 6.4 (grading law). *For every even m and N large enough, $S_m = c_m N^{-(2m-1)}(1 + O(N^{-2}))$.*

The computed values are $c_2 = -4$, $c_4 = -32$, $c_6 = -1748/3$, $c_8 = -123332/9$ and $c_{10} = -49593808/135$. Corollary 6.3 gives the exponent $m + 1$, which coincides with $2m - 1$ only at $m = 2$. Writing the centre-of-mass term as $-\lambda P^2/N$, S_m is a polynomial in λ without constant term, and the computation shows that each coefficient of λ^j with $1 \leq j \leq m/2$ is separately $O(N^{-(2m-1)})$. What remains to be proved is that for $j < m/2$ the j -fold insertion of $(P^2 - 1)/N$, whose natural size is $N^{-(m-1)-2j}$, is in fact of this smaller order. The generic forms are valid only for $N \geq m + 3$; at smaller rank determinant effects of the kind in Theorem 4.1 enter, and for instance the generic form of S_4 has a double pole at $N = 3$.

Remark 6.5 (the same pattern in the band). The closed forms of Sections 3 and 5 show the same suppression for the full lattice problem: the second-order channels are $O(N^{-1})$ and $t_N = O(N^{-3})$; the fourth-order channel terms are $O(N^{-3})$ and $\alpha_N, \beta_N = O(N^{-7})$. In terms of the 't Hooft coupling $\lambda_t = g^2 N$ one has $u = N^2/\lambda_t^2$, so a coefficient of order $N^{-(4k-1)}$ at order u^{2k} gives a term $N\lambda_t^{-4k}$ times a constant: every even order carries the same single power of N . Whether this persists at sixth order for the multi-plaquette clusters is open.

7 Verification

The script `verify_paper.py` distributed with this paper checks, in exact arithmetic: Proposition 3.4; the channel weights, (7) and t_N with its expansion; the parity of the link set T and the vertex (9) for $N \leq 11$; the symbols of Theorem 5.4 and the identity (22); the value ω_3 , the identity (19) at $N = 3, \dots, 200$, and the $SU(3)$ numbers; Lemma 5.8 by exact real-root isolation; and (30). The script `check_doublet_2N.py` compares (27) with a direct high-precision evaluation for $N = 3, \dots, 7$. The first script does not recompute the cumulants of Computed results 4.3, 5.2 and 5.5, which require the Wilson-loop engine; that code, its pinned outputs and the run records are available from the author. Of the statements in this paper, only the polynomial identity in (19) has been formalised in a proof assistant.

8 Open problems

1. Prove Conjecture 6.4; a Slater–Condon evaluation in the exact orbital basis of h reduces the $j = 1$ term to a product of two one-particle matrix elements between adjacent orbitals, which obey the same power law numerically.
2. Remove (A2) and (A3) by an independent enumeration of the fourth-order support at some $N \geq 5$, and recompute the corner cumulant there by a second method.
3. Determine whether R^2 occurs at sixth order, that is, whether the cubic tier appears with $D_{\text{shp}} = 3B_{\text{shp}}$.
4. Bound the orders $m \geq 5$ uniformly on a fibre, which would turn the fourth-order statements about the band into statements about H_{eff} .

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A Polynomials

$$\begin{aligned}
p_{14}(N) &= 16896N^{14} - 131616N^{12} + 451352N^{10} - 882908N^8 + 1058410N^6 \\
&\quad - 771029N^4 + 313093N^2 - 54216, \\
p_{18}(N) &= 2379648N^{18} - 28088736N^{16} + 143075272N^{14} - 411323454N^{12} + 732994774N^{10} \\
&\quad - 837251963N^8 + 611821212N^6 - 275614672N^4 + 69464470N^2 - 7465875, \\
R_{20}(z) &= (z-1)^3(2z-3)(2z-1)^3(3z-2)(3z-1)(4z-9)^3(4z-5)(4z-1) \\
&\quad \times (9z-25)(9z-16)(16z^2-44z+25)(16z^2-33z+16), \\
P_{17}(z) &= 2096187310080z^{17} - 45206560309248z^{16} + 448972002607104z^{15} \\
&\quad - 2723575470882816z^{14} + 11288692151812096z^{13} - 33888218411529728z^{12} \\
&\quad + 76218901019673664z^{11} - 131068691814847264z^{10} + 174326341061538992z^9 \\
&\quad - 180230597250871976z^8 + 144751635142984472z^7 - 89742150515602808z^6 \\
&\quad + 42388925672412712z^5 - 14916377727371552z^4 + 3768794520714128z^3 \\
&\quad - 641987460459360z^2 + 65414604672000z - 2967321600000.
\end{aligned}$$

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